

RESQNET AI: MASTER ENGINEERING SPECIFICATION

THE COMPLETE IMPLEMENTATION GUIDE, FALSE-POSITIVE PHYSICS FILTERS, AND RULEBOOK COMPLIANCE FRAMEWORK

Target Hackathon Topic: Topic 1.3 - RoadSoS (Emergency Accident Help System)

Host Body: Centre of Excellence for Road Safety (CoERS), RGB Labs, IIT Madras

Architecture Version: 2.5 (Production & Deployment Ready) | **Date:** May 2026

1. EXECUTIVE SUMMARY & CORE ENGINEERING PHILOSOPHY

During major high-impact vehicular collisions, the standard consumer mobile software ecosystem fails to deliver life-saving responses due to three critical infrastructure bottlenecks: immediate loss of network packets on un-networked highway segments, spatial information fragmentation, and full user cognitive/physical incapacitation (unconsciousness).

ResQNet AI addresses these limitations by shifting critical logic from cloud networks to local device hardware. This document outlines the absolute step-by-step framework to build a robust, zero-network, voice-first autonomous emergency assistant. It details the algorithms required to eliminate false positives from damaged infrastructure, outlines the compilation of localized embedded columnar databases, maps compliance rules against the strict grading rubrics of the IIT Madras evaluation committee, and configures the software for global scalability without code modifications.

2. DEEP-DIVE SYSTEM ARCHITECTURE & COMPONENT SELECTION

In accordance with the explicit mandates stated in the CoERS Handbook prioritizing open-source frameworks, local computational paradigms, and non-proprietary models, the structural blocks are organized as follows:

System Block	Selected Tech Components	Architectural Engineering Rationale
Telemetry Filtering	<code>Python 3.11</code> / <code>NumPy</code> Edge Loops	Processes high-frequency polling data from hardware accelerometers and gyroscopes at sub-millisecond rates.
Embedded Storage Engine	<code>DuckDB</code> + Spatial Flat Indexes	Executes compressed columnar storage inside native application processes. Avoids database network socket overheads.
Local Vocal Engine	<code>Vosk Acoustic Engine</code> / ONNX Runtime	Translates speech-to-text natively using compressed acoustic patterns needing zero external server calls. Memory footprints < 50MB RAM.
P2P Communication	<code>BLE 5.2 (Bluetooth Low Energy)</code> Stack	Uses native baseband radios to continuously broadcast asynchronous disaster tokens to nearby vehicular transceivers.
Geospatial Lookups	<code>OSRM</code> + Raw Protocol Buffers (<code>.pbf</code>)	Parses highly pruned and compressed geographic maps locally to index route constraints without relying on Google Maps APIs.

3. THE FALSE-POSITIVE MITIGATION PIPELINE (PHYSICS-DRIVEN FILTERING)

The primary operational risk for a vehicular crash application is user annoyance and unneeded emergency escalations driven by everyday road shocks, such as speed bumps, sudden aggressive braking, deep potholes, or dropping the device inside the cabin. ResQNet AI eliminates these false positives using a strict **Two-Stage Hardened Validation Pipeline**.

3.1 Stage 1: Mathematical Telemetry Splitting & Kinematic Analysis

The device continually streams telemetry from the internal tri-axial accelerometer and gyroscopic sensors at 50 Hz. The application computes the continuous aggregate Vector Magnitude Acceleration (A_{mag}):

$$A_{mag} = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$$

To differentiate a deep pothole or heavy braking from an actual crash event, the engine validates the signature against two tight kinematic thresholds simultaneously:

- **Kinematic Signature A (Everyday Environmental Noise):** Passing over deep potholes or emergency braking creates heavy acceleration readings up to **2.2G**. However, these spikes are highly transient, oscillating rapidly back to baseline equilibrium. The angular velocity vectors processed via the gyroscope indicate stable horizontal orientations. The software immediately discards these events.
- **Kinematic Signature B (True Collision Vector):** A high-impact collision registers an instantaneous deceleration spike exceeding **4.5G** within a strict temporal frame of $\Delta t \leq 120ms$. This is accompanied by sudden, structural alterations in gyroscopic angular changes exceeding **60°/sec** (indicating vehicle rollover or sudden kinetic dislocation). Only this signature triggers Stage 2.

3.2 Stage 2: Voice-First Persistence Loop & Fail-Safe Audio Window

If the mathematical criteria of Stage 1 are reached, the system avoids launching emergency network routines immediately. Instead, it enters an active conversational validation state designed to intercept false alarms natively:

1. **The 15-Second Voice Assessment Window:** The application opens the device's local audio capture subsystem and plays a clear verbal prompt: *"Potential collision vector identified. If you are safe, please say 'Cancel' or 'I am safe' immediately."*
2. **Continuous Edge Keyword Tokenization:** The local Vosk ASR engine listens actively for a predefined dictionary of local safety terms (e.g., *"I am safe"*, *"Cancel"*, *"Theek hu"*, *"No emergency"*). If any safety token matches the real-time transcript within this 15-second window, the system terminates the alert state, flushes the sensory queue, and returns to normal monitoring. No active button clicks are required.
3. **The 10-Second Physical Siren Escalation:** If the 15-second voice window passes with absolute silence or unparseable noise patterns (indicating the driver is unconscious or severely injured), the application transitions to its final verification step. It triggers a maximum-intensity haptic vibration and a loud audio siren for exactly 10 seconds. This localized warning serves to notify passing citizens and gives the user one last chance to tap a large screen override button if they are conscious but unable to speak.
4. **Autonomous Mode Transition:** If both validation windows (25 seconds total) pass with zero user input, the application declares a critical emergency state and launches autonomous operations.

4. EMBEDDED OFFLINE SPATIAL DATABASE & PROXIMITY SEARCH OPTIMIZATION

To meet the grading requirements of the handbook regarding "Reliability and Data Accuracy" and "Number of contacts fetched" under zero network conditions, ResQNet AI embeds a pre-loaded, compressed structural **DuckDB data instance** inside the client installation package.

4.1 Database Pruning, Spatial Sharding, and Indexing

To keep package sizes compact, regional map files (.pbf) obtained from OpenStreetMap are processed to remove non-essential entries, keeping only key safety layers:

- **Target Structural Schemas:** Emergency Trauma Units, Level-1 General Hospitals, Ambulance Stations, Towing Service Hubs, and Local Tire Outlets.
- **Geospatial Indexing via Flat Buffers:** All entries are pre-sorted into spatial coordinate bins using an offline grid configuration. When an emergency is triggered, the system extracts the current hardware GPS coordinates and performs a local mathematical calculation.

The Offline Proximity Query Architecture (Haversine Formula):

To rank the closest trauma facilities instantly without contacting Google/Apple Cloud APIs, compiled analytical queries calculate distances natively across the dataset:

$$d = 2R \cdot \arcsin(\sqrt{(\sin^2(\Delta\phi/2) + \cos(\phi_{\text{sub}1}) \cdot \cos(\phi_{\text{sub}2}) \cdot \sin^2(\Delta\lambda/2))})$$

Where ϕ and λ represent local latitude and longitude points, and R represents the Earth's radius. DuckDB handles this math using vector pipelines, sorting through thousands of records in less than 2 milliseconds.

4.2 Multi-Tiered Global Emergency Data Schema Layout

```
-- Complete Standardized Embedded Emergency Table Schema for Global Deployment
CREATE TABLE offline_emergency_facilities ( facility_id VARCHAR PRIMARY KEY,
iso_country_code VARCHAR(3) NOT NULL, -- Core Global Applicability Anchor
facility_name VARCHAR(150) NOT NULL, facility_type VARCHAR(30) CHECK(facility_type
IN ('TRAUMA_CTR', 'HOSPITAL', 'AMBULANCE', 'TOW_SERVICE', 'MECHANIC')), latitude
DOUBLE NOT NULL, longitude DOUBLE NOT NULL, primary_contact_number VARCHAR(20) NOT
NULL, alt_contact_number VARCHAR(20), triage_capability_level INT DEFAULT 1, -- 1
Basic, 2 Advanced, 3 Comprehensive administrative_district VARCHAR(50) ); -- Highly
Optimized Local Vector Search Query Execution Example CREATE INDEX spatial_geo_idx
ON offline_emergency_facilities (latitude, longitude);
```

5. GLOBAL APPLICABILITY ARCHITECTURE (MCC INTERCEPTION ENGINE)

The IIT Madras handbook requires "**Global applicability across countries**". Solutions that are locked to hardcoded Indian telephone protocols or specific local government backends will lose significant marks. ResQNet AI fulfills this requirement via an automated, hardware-aware country routing matrix.

5.1 SIM-Baseband Mobile Country Code (MCC) Isolation

On application startup, the software bypasses cellular IP checks and queries the device's native baseband radio interface directly to read the network operator's **Mobile Country Code (MCC)** parameters:

- **Dynamic Network Configuration Mapping:** The system matches the retrieved MCC string against an internal international configuration table. If the MCC matches India (**404** or **405**), the routing engine automatically assigns the primary local emergency dialer target to 112 and filters the DuckDB lookup context to target local administrative databases.
- **International Adaptation:** If the device moves internationally and registers an MCC from the United States (**310** or **311**), the backend dynamically adjusts the target emergency dialer string to 911, automatically remapping database queries to utilize the corresponding international hospital schemas.

This decouple-by-design method ensures that the identical software compilation can be launched across various regions worldwide with zero manual text modifications.

6. STEP-BY-STEP IMPLEMENTATION AND DIRECTORY CONSTRUCTION PLAN

To ensure your team develops a clean, professional solution within the limited hackathon timeline, the project layout must avoid unstructured single-file scripts. Use this production-ready directory structure and step-by-step assembly plan:

```

resqnet_ai_root/ | ├── README.md <-- System configuration instructions and
operational guide | ├── Assumptions.docx <-- Mandatory handbook documentation
outlining external variables | ├── requirements.txt <-- Locked version definitions for
dependency management | ├── database/ | ├── offline_facilities.db <-- Compiled
binary DuckDB columnar instance | ├── seed_data.csv <-- Raw historical seed files
utilized for verifying indexes | ├── src/ | ├── __init__.py | ├── sensors.py <-- High-
frequency hardware telemetry capture and filters | ├── database.py <-- Local DuckDB
connection controls and Haversine sorting | ├── triage.py <-- Local Vosk audio
streaming and keyword verification loops | ├── network_mesh.py <-- P2P BLE mesh
broadcast and receiver orchestration

```

6.1 Tactical Phased Development Schedule

1. **Phase 1: Local Storage Compilation:** Install the local DuckDB binaries. Load regional trauma center directories into the database and write the core SQL sorting scripts to ensure local query lookups complete in under 5 milliseconds.
2. **Phase 2: Kinematic Filtering Assembly:** Set up sensor polling mechanisms inside `sensors.py`. Code the mathematical Vector Magnitude filters to separate standard driving noise from sudden high-impact deceleration readings.
3. **Phase 3: Conversational Audio Integration:** Integrate the localized Vosk processing files. Test the voice validation loop to ensure the application resets cleanly upon parsing verbal safety commands without requiring touch interface inputs.
4. **Phase 4: Mesh Network Implementation:** Configure the device's BLE radio interfaces to continuous advertising modes, ensuring compressed data packets are broadcasted reliably during simulated zero-connectivity tests.

7. IIT MADRAS RULEBOOK COMPLIANCE & EVALUATION GRADING CRITERIA

Winning an IIT-hosted hackathon requires careful alignment with the evaluation parameters set by the organizing committee. Section 1.3.4 and Section 8.2 of the official guidelines outline strict requirements for submission validity.

7.1 Anti-Plagiarism Measures and MOSS Code Controls

The CoERS committee uses automated tools such as **MOSS (Measure of Software Similarity)** to evaluate incoming code blocks against global online code repositories and peer submissions. To prevent automatic rejection, ensure the following practices are followed:

- **Original Model Implementations:** Avoid copy-pasting code fragments from public web frameworks. Clearly write out your initialization sequences, input tensors, and local loop controls.
- **Comprehensive Code Comments:** Document the architectural rationale behind your functions, particularly explaining the local offline constraints, to demonstrate that the implementation was written entirely by your team.

7.2 Submission Payload Checklist for Portal Uploads

To pass the initial Stage 1 screening on the Unstop system, the final compressed submission archive must include:

1. **The Complete Codebase:** A well-structured folder structure containing functional code, free of dead code fragments or non-functional mock-ups.
2. **The Binary Database Layer:** The compiled, fully indexed local database storage files, proving that the solution functions reliably without calling cloud data servers.
3. **The 7-Slide Strategic Presentation:** A highly focused presentation deck matching the exact structural layout preferred by evaluation juries.
4. **The Supplementary Word Log (`Assumptions.docx`):** A clear text document mapping out all architectural assumptions, along with a complete list of open-source libraries used.

8. STRATEGIC PITCH DECK CONSTRUCTION (THE WINNING 7-SLIDE LAYOUT)

The handbook restricts the Stage 1 submission to exactly ****7 slides total****, including title and conclusion slides. To maximize impact, organize your pitch deck using this proven structure:

Slide No.	Slide Title	Mandatory Content and Key Focus Points
Slide 1	Title & Introduction	Project Name: ResQNet AI . Include team names, affiliation details, and an impactful project tagline highlighting the zero-network paradigm.
Slide 2	The Problem Landscape	Highlight the critical "Golden Hour" timeline. Explain how infrastructure gaps and driver shock cause communication failures during major highway accidents.
Slide 3	The ResQNet Solution	Present an overview of your edge-first model. Emphasize your key differentiators: hands-free voice operations, offline local databases, and zero-network availability.
Slide 4	Technical Architecture	Display a clean block diagram showing the data flow from physical sensors through the offline DuckDB instance to the local audio loops. Highlight the open-source tech stack.
Slide 5	False-Positive Filtering	Detail your two-stage mathematical and verbal verification engine. Prove how your solution avoids false alarms from rough roads while maintaining safety for unconscious drivers.
Slide 6	Global Scalability	Explain the MCC baseband tracking architecture. Show how the database schema scales globally across multiple countries without code refactoring.
Slide 7	Feasibility & Vision	Outline your implementation roadmap and deployment strategies with transport ministries (MoRTH/NHAI). Conclude with a clear summary slide.